FOOD RECOMMENDATION SYSTEM

Project Report

AYUSH SHUKLA
25BAI10014
VIT BHOPAL UNIVERSITY
FUNDAMENTALS IN AI & ML

1. Introduction

In today's fast-paced world, people often struggle to decide what to eat. A **Food Recommendation System** helps users choose meals based on their preferences, mood, and time of day.

This project is a simple yet effective application developed using Python and Tkinter that provides personalized food suggestions. It demonstrates how recommendation systems work using rule-based logic.

🎯 2. Objective

The main objectives of this project are:

- To develop a user-friendly GUI-based application
- To recommend food items based on user preferences
- To understand basic concepts of recommendation systems
- To implement Python programming with Tkinter
- To enhance user interaction through a modern UI

🛠 3. Technology Used

Technology	Description
Python	Core programming language
Tkinter	GUI (Graphical User Interface) library
Random Module	Used for random food selection

⚙ 4. System Requirements

Hardware Requirements

- Computer/Laptop
- Minimum 4GB RAM

Software Requirements

- Python 3.x
- Any Code Editor (VS Code / PyCharm / IDLE)

□ 5. Working Principle

The system works based on **rule-based filtering**:

1. The user selects:
    - Diet type (Veg/Non-Veg)
    - Meal type (Breakfast/Lunch/Dinner)
    - Mood (Happy/Sad/Stressed/Lazy)
2. The system processes the input and:
    - Matches it with predefined food data
    - Adds mood-based suggestions
    - Uses random selection to give variety
3. The result is displayed with a typing animation.

□ 6. System Design

Input:

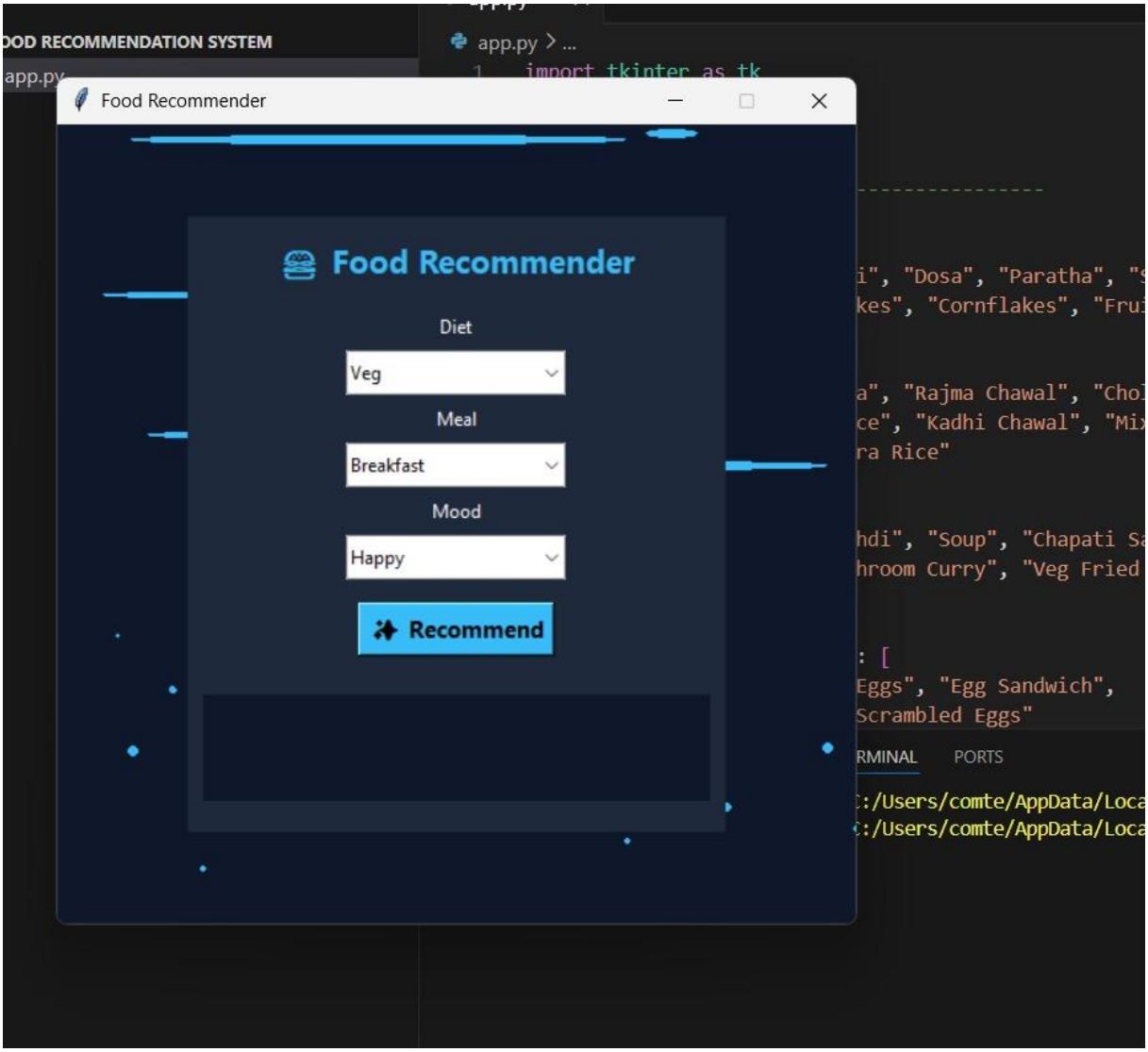
- User selections from dropdown menus

Process:

- Matching conditions using dictionaries
- Combining multiple lists
- Random selection algorithm

Output:

- Suggested food item displayed on screen



🔄 7. Algorithm

1. Start
2. Take user input (diet, meal, mood)
3. Check matching food category
4. Add mood-based items
5. Combine all suggestions
6. Select a random food item
7. Display result

8. Stop

💻 8. Implementation

The project is implemented using Python with Tkinter GUI. It includes:

- Dropdown menus (Combobox)
- Buttons for interaction
- Canvas for animated background
- Labels for displaying results

The application also includes animation features like:

- Moving particles background
- Typing effect for output

✨ 9. Features

- User-friendly interface
- Dark theme modern UI
- Animated background
- Smart recommendation logic
- Variety of food items
- Fast and responsive system

📊 10. Advantages

- Easy to use
- Helps users decide quickly
- Lightweight application
- No internet required
- Beginner-friendly implementation

⚠️ 11. Limitations

- Not AI-based (rule-based only)
- Limited dataset
- No user history tracking
- No real-time data integration

🚀 12. Future Scope

- Integration with Machine Learning
- Personalized user profiles
- Food calorie tracking
- Voice-based input system
- Web or mobile application version

14. Conclusion

The Food Recommendation System successfully demonstrates how user input can be used to provide personalized suggestions. It is a simple yet powerful project that combines GUI design with logical programming.

This project also serves as a foundation for more advanced AI-based recommendation systems.

---

15. References

- Python Official Documentation
- Tkinter Documentation
- Online tutorials and resource

Acknowledgement

I would like to thank my teacher and institution for providing me the opportunity to work on this project. I also thank online resources and documentation that helped me complete this project successfully.